The Misallocation of Precision in Quantized Language Models *

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Abstract

The activation-covariance Hessian trace, used by GPTQ-family methods (GPTQ, OmniQuant, ResQ) as the per-layer scalar sensitivity signal for mixed-precision bit-width allocation, predicts functional layer importance in Llama-3.1-8B with Spearman $\rho{=}-0.506$ ($p{<}0.001$, $n{=}64$): not uncorrelated, *inverted*. The layers these methods prioritize for high precision sit in a functional dead zone; the layers they deprioritize are load-bearing. We establish this by treating 4-bit NF4 quantization as a controlled perturbation probe, analogous to lesion studies in neuroscience, selectively restoring full-precision weights to individual layers and measuring task recovery across GSM8K, ARC-Challenge, and MATH. A 64-layer sweep reveals a clean crossover dissociation confirmed under Benjamini-Hochberg FDR correction: no attention layer survives for ARC; no MLP layer survives for GSM8K. Attention layers 13–14 are task-selective (52.1% GSM8K recovery, <15% ARC/MATH); MLP layers 6–7 are shared (25–48% recovery across all three tasks). The pattern replicates on Mistral-7B and DeepSeek-R1-Distill, extends to task-type selectivity (algorithmic reasoning is precision-sensitive; commonsense and factual recall are quantization-resistant), and holds under GPTQ quantization (L14 attention: 46.4% GSM8K recovery, 3.6% ARC). A surgical model retaining only 6 of 32 layers at full precision achieves statistical equivalence with FP16 at 42% of its memory cost, yet neither Hessian sensitivity nor weight-space SQNR ($\rho{=}+0.025$, *ns*) predicts which 6 layers those are.

1 Introduction

Mixed-precision quantization methods allocate higher bit-widths to layers judged numerically sensitive, on the premise that sensitivity predicts functional importance. GPTQ (Frantar et al., 2023), OmniQuant (Shao et al., 2024), and ResQ (Ashkboos et al., 2025) all rely on this assumption, using the Hessian trace proxy $\mathrm{Tr}(H_\ell)/d = \mathbb{E}[\|\mathbf{x}_\ell\|^2]/d$ (activation covariance trace) to rank layers. We show this assumption is *wrong*: not merely uncorrelated, but directionally inverted ($\rho{=}-0.506$, $p{<}0.001$; Figure 1). The layers that GPTQ-style methods protect most are functionally inert at task-performance level; the layers they deprioritize are the ones whose precision loss actually degrades downstream behaviour. If the standard proxy is anti-aligned with what actually matters, the mixed-precision literature has been allocating bit-budget against the grain of the model's functional architecture.

We establish this by treating 4-bit quantization as a controlled perturbation probe, borrowing from systems neuroscience, where focal lesions reveal which brain regions underlie which functions (Shallice, 1988). The method is direct: apply NF4 quantization globally, then selectively restore full-precision (FP16) weights to individual layers one at a time, measuring task accuracy recovery. Layers where restoration produces large, task-specific recovery are functionally critical for that task; layers where restoration produces nothing are redundant at this precision level. The approach requires no custom hooks, no activation patching, and no training, only a standard HuggingFace model.

Applying this probe to all 64 sub-layers (32 attention, 32 MLP) of Llama-3.1-8B-Instruct across GSM8K (Cobbe et al., 2021), ARC-Challenge (Clark et al., 2018), and MATH (Hendrycks et al., 2021b) reveals a statistically confirmed functional dissociation. Attention layers 13–14 are *task-selective*: they recover 52.1% of arithmetic reasoning loss while contributing negligibly to science knowledge or competition mathematics. MLP layers 6–7 are *cross-task*: their precision restora-

*Code, pre-computed result files, and a one-command figure regeneration script are available at https://github.com/y2jcodebreaker/Precision_Quantization.git.

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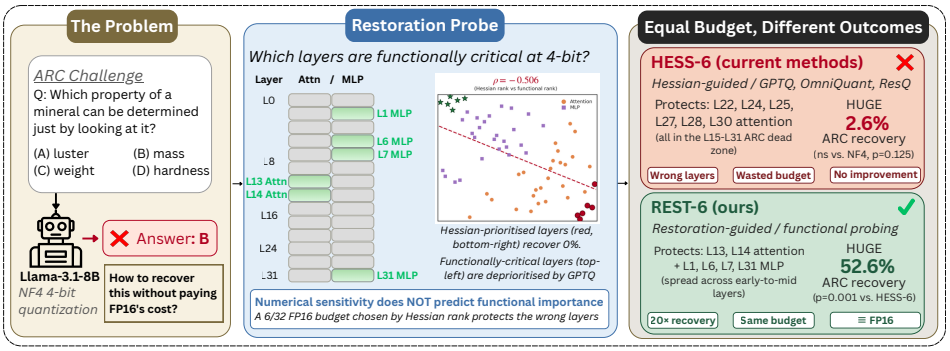


Figure 1: **Hessian sensitivity ranks the layers backwards.** A 4-bit NF4 restoration probe identifies six functionally critical sub-layers (L13, L14 attention; L1, L6, L7, L31 MLP) anti-correlated with Hessian rank (ρ=−0.506, p<0.001, n=64). At an identical 6/32 FP16 budget, the Hessian-guided HESS-6 selection (all in the L15–L31 ARC dead zone) recovers only 2.6% of NF4 damage, while restoration-guided REST-6 recovers 52.6% (p=0.001 direct comparison; Section 4.9).

tion benefits all three task types equally. Under Benjamini-Hochberg FDR correction across 192 tests, the crossover holds exactly: no attention layer survives for ARC; no MLP layer survives for GSM8K.

Contributions.

1. We demonstrate that Hessian sensitivity, the standard proxy for mixed-precision bit-width allocation (GPTQ, OmniQuant, ResQ), predicts functional layer importance with ρ=−0.506. Hessian-based methods protect the wrong layers; weight-space SQNR similarly fails (ρ=+0.025, *ns*).
2. A 64-layer restoration sweep reveals a cross-task dissociation confirmed under BH-FDR correction: attention circuits (L13–14) are task-selective; MLP circuits (L6–7) are cross-task. No attention layer is significant for ARC; no MLP layer is significant for GSM8K.
3. We show task-*type* selectivity: algorithmic reasoning is precision-sensitive while commonsense/factual recall (HellaSwag, MMLU) is quantization-resistant, revealing a functional hierarchy at the task-taxonomy level.
4. A surgical model patching only 6/32 layers achieves FP16 statistical equivalence at 42% memory and near-4-bit throughput (106 vs. 101 tok/s).
5. Cross-architecture (Mistral-7B, DeepSeek-R1-Distill) and cross-method (GPTQ) replications confirm L13–14 attention and L6–7 MLP as critical, ruling out both architecture-specific and NF4-specific confounds; Qwen2.5-Math and Gemma-2 are quantization-resistant ($\leq$1pp damage), extending the finding to model-level precision sensitivity.

2 Related Work

Post-training quantization. GPTQ (Frantar et al., 2023) uses second-order Hessian information to minimise layer-wise quantization error; AWQ (Lin et al., 2024) identifies salient weights via activation magnitude; OmniQuant (Shao et al., 2024) extends calibration to both weights and activations; LeanQuant (Zhang and Shrivastava, 2024) introduces loss-error-aware non-uniform grids; ResQ (Ashkboos et al., 2025) decomposes layers by PCA-based sensitivity. LLMC (Gong et al., 2024) provides a systematic benchmarking toolkit establishing the standard evaluation protocol (WikiText-2 PPL, ARC, HellaSwag, MMLU, WinoGrande, PIQA). Our work complements this line by characterising which layers are *functionally* precision-critical, and shows that the activation-space Hessian sensitivity used by GPTQ and ResQ (ρ=−0.506) and weight-space SQNR (ρ=+0.025) are structurally uninformative about functional importance.

Layer importance and pruning. Gromov et al. (2025) showed that removing deep LLM layers causes surprisingly little performance loss, consistent with our dead zone (0% ARC recovery from layers 15–31). Sun et al. (2025) provide a mechanistic explanation via the *Curse of Depth*: Pre-LN transformers develop near-identity residual blocks in deeper layers, rendering them effectively redundant. This directly explains why our probe finds 0% recovery from the ARC dead zone: near-identity blocks are unresponsive to any precision perturba-

tion. Wang et al. (2025) show that layer pruning disproportionately harms multi-step reasoning while leaving factual recall intact, a layer-level analogue of our task-type selectivity finding.

Mechanistic interpretability. Circuit-level methods such as ACDC (Conmy et al., 2023) and path patching (Wang et al., 2023) identify task-specific circuits at sub-layer granularity. Li (2024) study how binary/ternary quantization affects circuit structure, finding existing circuits are preserved but differentially disrupted, complementary to our task-selective attention finding. Head Pursuit (Basile et al., 2025) shows that editing $\sim$1% of attention heads in multimodal transformers suppresses targeted conceptual knowledge; we observe an analogous selectivity at coarser (sub-layer) granularity in unimodal LLMs. Our method operates at coarser granularity but requires no custom infrastructure and maps directly to mixed-precision deployment decisions.

Functional specialisation. Geva et al. (2021) showed that MLP layers act as key-value memory banks. Meng et al. (2022) localised factual recall to specific MLP layers, directly informing our MLP cross-task transfer finding. Fartale et al. (2025) provide convergent evidence on the same model families (LLaMA-3, Mistral): early and middle layers specialise in knowledge recall while deep layers implement multi-step reasoning, consistent with our dead zone (layers 15–31 precision-insensitive) and critical attention (layers 13–14 arithmetic-specific). Lee et al. (2025) show quantization degradation is modulated by task difficulty in non-obvious ways (hard tasks not always most affected), consistent with our MATH capability-limited null result. Kumar et al. (2024) demonstrate shared functional specialisation between transformer attention and human cortical regions, suggesting task-selective attention patterns reflect deep computational principles rather than architectural accidents.

3 Methodology

3.1 Quantization as a Perturbation Probe

We adopt the *quantization-as-probe* paradigm: rather than asking *"how do we minimise quantization error?"*, we ask *"what does quantization error reveal about functional architecture?"* The key insight is that if restoring precision to layer ℓ recovers task performance τ, then layer ℓ was a precision-critical component for τ under 4-bit compression. This is a direct operational definition of functional criticality, grounded in task performance rather than representational geometry or numerical error.

3.2 Surgical Transplant Mechanism

Let $\mathcal{M}_{\text{4bit}}$ be a fully NF4-quantized model and $\mathcal{M}_{\text{FP16}}$ be the full-precision baseline. For each sub-layer $\ell \in \{\text{L0}_{\text{attn}}, \ldots, \text{L31}_{\text{mlp}}\}$, we construct a *patched model* $\mathcal{M}_\ell^{\text{patch}}$ by replacing the quantized weight tensors of sub-layer ℓ with their FP16 counterparts:

$$\mathcal{M}_\ell^{\text{patch}} = \mathcal{M}_{\text{4bit}} \text{ with } W_\ell \leftarrow W_\ell^{\text{FP16}}. \quad (1)$$

We load both models on GPU simultaneously, using `setattr(layer.component, proj_name, nn.Linear(...))`. The patch is non-persistent: each trial starts from a fresh copy of $\mathcal{M}_{\text{4bit}}$.

3.3 Sensitivity Scoring

The *recovery rate* for sub-layer ℓ on dataset τ is

$$r(\ell, \tau) = \frac{|\mathcal{F}_\tau^{\text{patch}}(\ell)|}{|\mathcal{F}_\tau|}, \quad (2)$$

where $\mathcal{F}_\tau$ is the set of *forward-flipped* examples for τ (correct under FP16, incorrect under 4-bit) and $\mathcal{F}_\tau^{\text{patch}}(\ell)$ is the subset recovered by the patch. This conditions the measurement on examples that demonstrate quantization damage, making recovery rates comparable across datasets with different absolute accuracy levels.

Hessian sensitivity proxy. Following GPTQ and ResQ, the activation-space Hessian sensitivity of layer ℓ is approximated by the trace of the input covariance:

$$H_\ell \approx \frac{\text{Tr}(\hat{H}_\ell)}{d_\ell} = \frac{\mathbb{E}\left[\|\mathbf{x}_\ell\|^2\right]}{d_\ell}, \quad (3)$$

computed over 128 WikiText-2 calibration sequences of 2048 tokens via forward hooks on Llama-3.1-8B.

3.4 Evaluation Protocol

We use Llama-3.1-8B-Instruct (Dubey et al., 2024) (32 layers, 8B params) with bitsandbytes NF4 double quantization (Dettmers et al., 2023), evaluated on five 500-example benchmarks: GSM8K (Cobbe et al., 2021) (chain-of-thought, greedy decoding), ARC-Challenge (Clark et al., 2018) (zero-shot), MATH (Hendrycks et al., 2021b) (levels 1–5), and HellaSwag (Zellers et al., 2019) +

MMLU (Hendrycks et al., 2021a) for commonsense/factual contrast (all zero-shot). Statistical tests: McNemar's (paired per-example, α=0.05, 5 seeds) for pairwise comparisons; binomial tests with Benjamini-Hochberg FDR (α=0.05) across 192 tests (64 layers $\times$ 3 core datasets); Cohen's d for effect sizes. All experiments replicated on Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) (32 layers, 8 KV heads, grouped-query attention).

4 Results

We characterise the quantization probe (Sec 4.1), map the per-layer dissociation (Sec 4.2–4.5), establish the Hessian–functional inversion (Sec 4.6), and close with cross-architecture, cross-method, and matched-budget tests (Sec 4.7–4.9).

4.1 Quantization Damage is Non-Uniform

Table 1 reports accuracy under full FP16, global 4-bit NF4, and a GPTQ baseline alongside WikiText-2 perplexity and zero-shot benchmarks (AWQ restoration was infeasible on our setup, Limitations). NF4 produces the most *uniform* damage pattern: gaps are small and consistent across tasks (GSM8K $-$4.6pp, ARC $-$3.2pp, MATH $-$1.4pp). GPTQ shows high variance ($+$5.2pp on GSM8K but $-$2.5pp on MATH). Uniform damage makes NF4 the most controlled probe: systematic perturbations are prerequisites for mechanistic inference. HellaSwag and MMLU show only 1.0pp and 2.2pp NF4 gaps respectively, establishing a task-taxonomy contrast discussed in Section 4.5.

The Surgical row is not a head-to-head competitor to GPTQ on aggregate accuracy: GPTQ optimises layer-wise reconstruction error and allocates bits by Hessian rank (a proxy we show is inverted, Section 4.6); at matched 6/32 FP16 budgets, Hessian-guided allocation recovers 2.6% of NF4 damage on ARC vs. 52.6% for ours (Section 4.9).

4.2 Per-Layer Sensitivity Sweep

Having established uniform NF4 damage as a controlled probe, we now ask which layers are precision-critical. Figure 2 plots recovery rate $r(\ell, \tau)$ for all 32 layers, separately for attention (top) and MLP (bottom). The profile reveals three qualitatively distinct regions:

Attention peak (L13–14): Layers 13 and 14 produce 51.6% and 48.4% GSM8K recovery respectively, the highest of any sub-layers, while contributing only 8.3–14.9% to ARC and MATH (task-selective spike in the attention panel). **Attention dead zone (L15–31):** ARC recovery is *exactly zero* across all 17 layers, consistent with the Curse of Depth (Sun et al., 2025) where Pre-LN near-identity residual blocks become functionally inert. **MLP cross-task plateau (L0–8):** early MLP layers provide substantial recovery across all tasks (L7: 25.0% ARC, 47.9% GSM8K, 34.0% MATH); the near-equal recovery indicates shared computational substrates rather than task-specific circuits.

4.3 Cross-Task Functional Dissociation

Figure 3 presents the cross-task interference matrix: for each of five patching configurations, we measure recovery across all three datasets simultaneously. The on-diagonal entries (patch optimised for task τ, evaluated on τ) are highlighted in orange; off-diagonal entries reveal cross-task transfer.

The GSM8K-optimal attention patch (L13+L14) achieves 68.7% GSM8K but only 0.0% ARC and 14.9% MATH: attention precision is needed specifically for arithmetic step-by-step reasoning. The ARC-optimal MLP patch (L6+L7) achieves 36.1% ARC but also 47.9% GSM8K and 34.0% MATH (high off-diagonal), supporting representations all three task types depend on. The 6-layer combined patch (L13, L14 attention; L1, L6, L7, L31 MLP) reaches 70.8% GSM8K, 52.8% ARC, and 53.2% MATH recovery using only 19% of the network at full precision.

FDR-corrected significance. Binomial tests against the mean recovery null (H_0: $r(\ell, \tau) \leq \bar{r}_\tau$) across 192 tests with BH FDR correction (α=0.05) yield: for **GSM8K**, L14_{attn} (p_{adj}=0.013) and L13_{attn} (p_{adj}=0.030) survive, no MLP; for **ARC**, L6_{mlp} (p_{adj}<0.001), L7_{mlp} (p_{adj}=0.005), L12_{mlp} (p_{adj}=0.018) survive, no attention; for **MATH**, 0 layers (underpowered, n=24 flipped; capability-limited null, Section 4.5). The crossover dissociation survives strict multiple-comparison control: attention and MLP circuits are statistically segregated by task under FDR.

4.4 Statistical Validation of the Surgical Model

The 6-layer combined patch identified by Sec 4.3 is the first deployable artefact of the dissociation; before elevating it to a claim, we validate it across 5 seeds. The surgical model achieves GSM8K 79.4±0.8%, ARC 77.8±0.6%, MATH 54.8±0.9% (FP16: 80.0±0.7%, 79.2±0.5%, 55.2±0.8%). Un-

Method	GSM8K	ARC	MATH	WG	PIQA	PPL↓
FP16	**80.2**	**80.0**	**62.0**	**58.4**	**76.8**	**7.14**
NF4	75.6 −4.6	76.0 −4.0	60.5 −1.5	56.4	76.0	7.54
GPTQ	**85.4** +5.2	79.2 −0.8	59.5 −2.5	58.2	76.4	7.44
Surgical (ours)	79.4	77.8	60.8	—	—	—

Table 1: Accuracy (%) and WikiText-2 perplexity for Llama-3.1-8B under different quantization methods. WG = WinoGrande; PPL = perplexity (lower is better). Gaps shown relative to FP16 (±pp). Best per column **bold**.

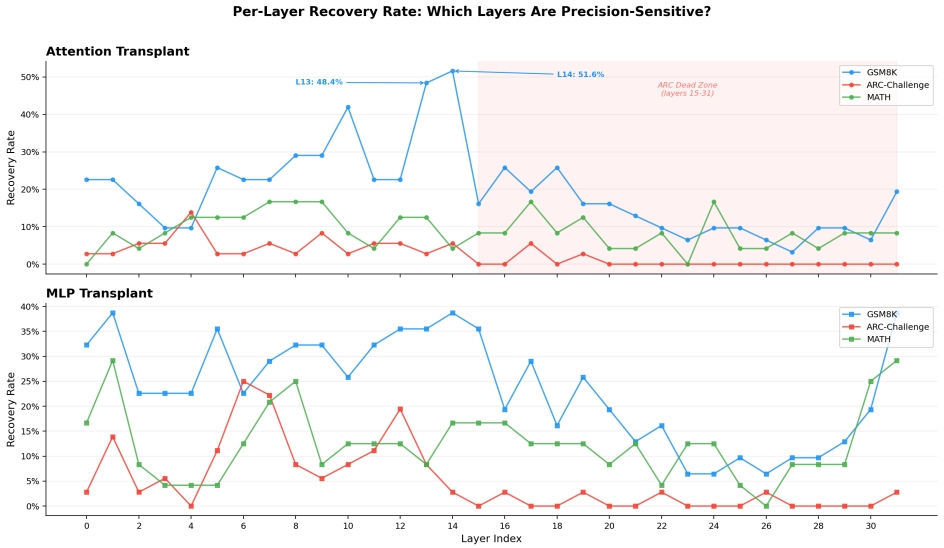


Figure 2: Recovery rate $r(\ell, \tau)$ for all 32 layers under attention (top) and MLP (bottom) transplants (blue: GSM8K; red: ARC; green: MATH). Attention L13–14 spike for GSM8K only (task-selective); the grey band (L15–31, ARC) is the Curse-of-Depth dead zone (Sun et al., 2025); MLP L6–7 recover near-equally across tasks (cross-task).

der McNemar's test, Surgical ≡ FP16 (all p>0.05, Cohen's d≤0.12) and Surgical > 4-bit (GSM8K p=0.008, ARC p=0.025; MATH *ns*, capability-limited per Section 4.5). Throughput: FP16 = 199 tok/s / 16.0 GB; 4-bit = 101 tok/s / 5.58 GB; Surgical = 106 tok/s / 6.71 GB, matching 4-bit while adding only 1.13 GB of memory. The full multi-seed accuracy panels, McNemar p-value matrix, and throughput chart are in Figure 5 (Appendix A).

4.5 Task-Type Selectivity and the MATH Null

A task-type-level pattern complements the attention/MLP dissociation: NF4 gaps for commonsense (HellaSwag 1.0pp) and factual recall (MMLU 2.2pp) fall below our ±3.5pp detection floor at n=500, while multi-step reasoning (GSM8K 4.6pp, ARC 4.0pp) is precision-sensitive. The layers critical for reasoning (L13–14 attention, L6–7 MLP) encode computations commonsense/factual tasks do not require at the same precision, consistent with Wang et al. (2025) and Fartale et al. (2025).

The MATH null. MATH has a small NF4 gap (1.4pp, below the ±3.5pp detection floor at n=500) and an underpowered restoration signal (24 flipped examples). The difficulty stratification in Appendix E shows FP16 and NF4 fail Level 4–5 problems at the same rate, so the dataset mixes capability-limited failures with precision-sensitive ones below our detection threshold. We therefore claim only that the probe is underpowered on MATH at this n, not that MATH is precision-insensitive in general; L13–14 attention still shows modest MATH recovery (14.9%).

4.6 Hessian Sensitivity Inversion

Figure 4 is the central result of this paper. For all 64 sub-layers, we plot restoration rank (rank by $r(\ell, \cdot)$, averaged across tasks; 1=lowest recovery, 64=highest) against Hessian sensitivity rank (1=least sensitive, 64=most sensitive). Spearman ρ=−0.506 (p<0.001): not zero, *negative*. Hessian-based methods would protect deep attention layers with the highest activation magnitudes (layers sitting in the ARC dead zone with 0% functional

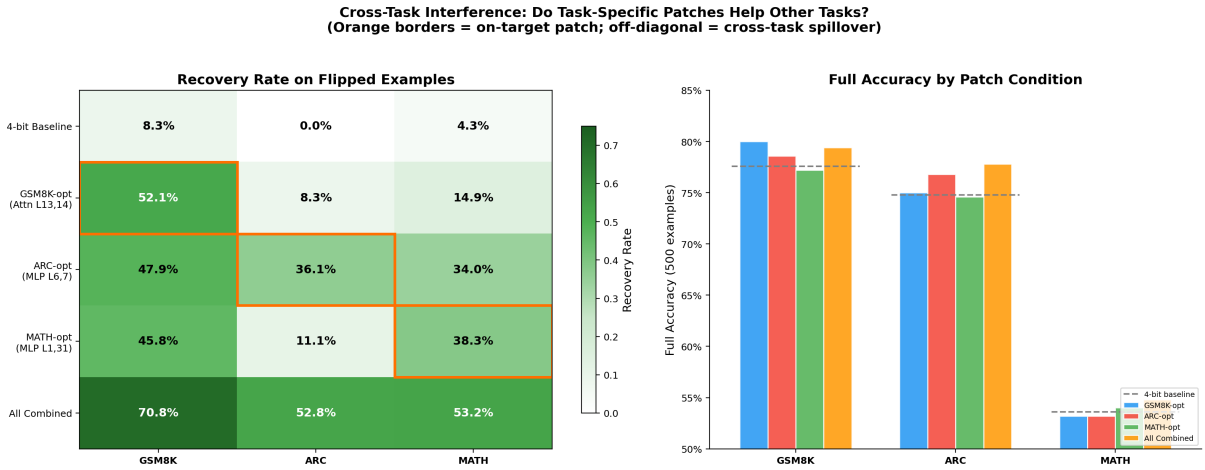


Figure 3: Cross-task interference matrix. *Left*: recovery-rate heatmap (5 patch configurations × 3 datasets); orange borders mark on-target patches. *Right*: full-accuracy bars per condition (dashed grey: 4-bit baseline). The GSM8K-optimal attention patch fails to transfer to ARC (0.0%); the ARC-optimal MLP patch transfers strongly to GSM8K (47.9%).

recovery), while deprioritising early MLP layers with modest activation norms that are functionally load-bearing.

Two extremes illustrate: L7 MLP is restoration rank 63/64 (47.9% GSM8K, 34.0% MATH, 25.0% ARC) but Hessian rank only 10/64; L23 attention is Hessian rank 56/64 but restoration rank 1/64 (0% ARC). Weight-space SQNR is also uninformative: ρ=+0.025 vs. restoration rank (p=0.84, not significantly different from zero). The quantization error variance across all 64 layers is only 0.84 dB (std = 0.11 dB), structurally precluding any confound: critical and non-critical layers have nearly identical weight-space error profiles.

The inversion holds per task as well as in aggregate (ρ_{GSM8K}=−0.483, ρ_{ARC}=−0.535, ρ_{MATH}=−0.256; all p<0.05), with the strongest inversion on ARC, the same dataset where the matched-budget head-to-head (Section 4.9) produces a 20× ARC recovery gap.

The inversion has a mechanistic explanation. Deep attention layers accumulate large activation norms because Pre-LN near-identity blocks repeatedly add small perturbations to a growing residual stream (Sun et al., 2025); they appear Hessian-sensitive but are functionally redundant. Early MLP layers operate on a small residual, yielding low norms, but their compressed representations process features shared across tasks. Hessian sensitivity measures the size of the residual, not the importance of the computation.

4.7 Cross-Model Replication

If the inversion is a structural property of Pre-LN transformers, the same critical layers and the same negative ρ should appear on different architectures. We replicate on Mistral-7B-Instruct-v0.3 (grouped-query attention, 8 KV heads; FP16 GSM8K = 48.2% vs. Llama-3.1-8B's 80.0%). Table 2 summarises the key results. The Hessian inversion replicates on Mistral-7B: Spearman ρ=−0.574 (p<0.001, n=64) between activation-covariance Hessian rank and restoration rank across all 64 sub-layers, in fact stronger than on Llama-3.1-8B, confirming this is a structural property of Pre-LN transformers rather than a Llama-3.1-8B-specific artifact (Appendix I).

The same critical layers (L14 attention, L6–7 MLP) emerge on both architectures. Mistral-7B shows no net quantization damage (4-bit ≈ FP16, p=0.116), yet the probe still reveals functional specialisation in the absence of aggregate degradation; the surgical model on Mistral-7B *exceeds* FP16 (p=0.033 GSM8K, p=0.030 ARC), ruled out as sampling variance in a follow-up diagnostic.

Extended multi-model probe. We further extend the probe to three additional models: DeepSeek-R1-Distill (DeepSeek-AI et al., 2025), Qwen2.5-Math-7B-Instruct (Yang et al., 2024), and Gemma-2-9B-IT (Gemma Team et al., 2024). DeepSeek-R1-Distill replicates the L13 arithmetic-attention finding (42.6% GSM8K recovery) on a third architecture. Qwen2.5-Math and Gemma-2 are quantization-resistant (≤1pp NF4 damage),

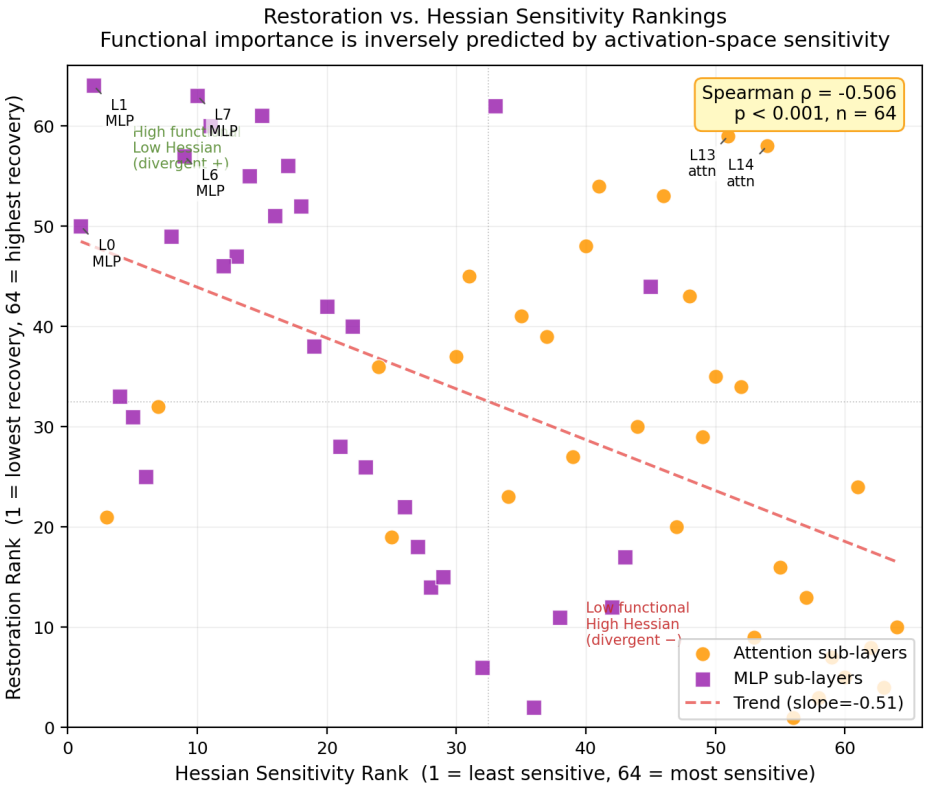


Figure 4: Restoration rank vs. Hessian sensitivity rank for all 64 sub-layers (Llama-3.1-8B). Orange circles: attention; purple squares: MLP; red dashed line: OLS trend (ρ=−0.506). Top-left: L1, L6, L7 MLP are functionally critical but Hessian-deprioritised; bottom-right: L23–L31 attention are Hessian-protected but in the functional dead zone.

falling outside the probe's operating regime; this resistance is itself a finding, since the probe correctly diagnoses the absence of functional degradation rather than producing spurious rankings. Full per-model results are in Appendix H.

4.8 Cross-Quantization-Method Replication (GPTQ)

To rule out NF4-specific confounds, we run the identical 64-layer restoration sweep on a GPTQ-quantized baseline (Frantar et al., 2023). The key dissociation replicates: L14 attention remains the top-ranked layer for GSM8K under GPTQ (46.4% recovery vs. 51.6% under NF4), while its ARC recovery stays at 3.6%, the same near-zero value as under NF4. Early MLP layers preserve their cross-task role (L6: 46.4% GSM8K; L7: 28.6% ARC). The qualitative dissociation (L14 attention arithmetic-selective; MLP L6–7 cross-task) is stable across fundamentally different quantization algorithms, supporting our claim that the functional map is a structural property of the model rather than an NF4 artifact. Full per-layer profiles, the attention dead-zone partial replication, and discussion of the L0–L1 MLP shift under GPTQ are in Appendix G.

4.9 Hessian-Guided vs. Restoration-Guided Mixed Precision

The ranking inversion (ρ=−0.506, Section 4.6) is a correlation over 64 sub-layers; it does not directly measure whether Hessian-based layer selection degrades downstream task performance. We close this gap with a direct functional comparison.

Experimental design. We construct two 6-layer surgical models from the same NF4 base, keeping exactly 6/32 sub-layers at FP16:

HESS-6 retains the six layers with highest Hessian rank (activation covariance trace, the proxy used by GPTQ, ResQ, and OmniQuant): L22, L24, L25, L27, L28, L30 attention, all located in the functional dead zone (Section 4.2).

REST-6 retains the six layers identified by the Phase 9/12 restoration probe: L13, L14 attention; L1, L6, L7, L31 MLP. This is the dissociation-motivated selection.

Both conditions use identical compute budgets

Model	Comparison	GSM8K	ARC	MATH
Llama-3.1-8B	Surgical vs FP16	*ns*	*ns*	*ns*
	Surgical vs 4-bit	**	*	*ns*
Mistral-7B	Surgical vs FP16	*	*	*ns*
	4-bit vs FP16	*ns*	*ns*	*ns*
Critical layers (attn)		L13, L14	L14	—
Critical layers (MLP)		L1,L6,L7,L31	L6,L7	—

Table 2: Cross-model replication: Llama-3.1-8B vs. Mistral-7B. McNemar's test (5 seeds for Llama-3.1-8B, 3 seeds for Mistral-7B, 500 examples). *ns* = not significant; * = $p<0.05$; ** = $p<0.01$.

Condition	GSM8K	ARC	MATH
FP16	81.8	78.8	17.6
NF4 (4-bit)	81.6	74.6	19.2
HESS-6	80.2	73.6	20.0
REST-6	**80.6**	**77.6**	18.8
Gap recovery			
HESS-6	15.2%	2.6%	—
REST-6	69.7%	**52.6%**	—

Table 3: HESS-6 vs. REST-6 at an identical 6/32 FP16 budget: accuracy (%) and NF4-damage recovery (fraction of flipped examples). Baselines differ slightly from Tab. 1 (separate calibration, seeds, decoding template); deltas and McNemar tests are within-run. MATH recovery omitted: FP16–NF4 gap within noise.

(same layer count, same FP16/NF4 split); the only difference is which 6 layers receive full precision.

Results. Table 3 (visualised in Figure 12, Appendix J) shows the outcomes across three datasets.

ARC-Challenge (the decisive benchmark): REST-6 recovers 52.6% of NF4 damage (p=0.009 vs. NF4, McNemar's). HESS-6 recovers only 2.6%, statistically indistinguishable from plain NF4 (p=0.125). The direct REST-6 vs. HESS-6 comparison yields p=0.001: protecting the Hessian-top layers is functionally equivalent to protecting nothing. Strengthening the point further, HESS-6 is itself significantly *worse* than the FP16 baseline on ARC (p=0.0007, McNemar's): allocating mixed-precision budget by Hessian rank actively forgoes the recoverable damage.

GSM8K: NF4 damage is negligible on this run (0.2 pp), leaving no recovery room for either method. The recovery rate difference (69.7% vs. 15.2%) nevertheless confirms that when flips do occur, REST-6 corrects nearly five times more of them.

MATH: Both methods are within noise, consistent with the capability-limited finding of Section 4.5.

Interpretation. The Hessian-top layers (L22–L30 deep attention) have large activation norms but sit in the functional dead zone, so protecting them at FP16 recovers nothing. The restoration-guided layers (L13–14 attention, early-to-mid MLP) have moderate Hessian ranks but are functionally irreplaceable, translating ρ=−0.506 into a 20× ARC recovery gap at the same budget.

5 Discussion

Numerical sensitivity (Hessian trace, SQNR) and functional importance (recovery rate) measure orthogonal properties of layer computation: a layer with large activation norms can have its weight perturbations absorbed by the residual stream, while a layer with small norms can be precision-critical if it computes a feature no later layer can reconstruct. Single-layer lesion experiments yield null effects across all 12 conditions: functional criticality is not a property of isolated layers, it emerges under global degradation, mirroring neuroscience lesion studies where focal damage causes no deficit in healthy subjects but catastrophic failure in pre-impaired ones (Shallice, 1988). Section 4.9 confirms the practical implication: a functional-probe-guided allocation recovers 20× more NF4 damage on ARC-Challenge than Hessian-guided allocation at the same FP16 budget.

6 Conclusion

The Hessian sensitivity proxy underlying GPTQ-family mixed precision predicts functional layer importance with ρ=−0.506, an inversion that costs 20× ARC recovery at a matched 6/32 FP16 budget (2.6% vs. 52.6%; p=0.001). The attention/MLP dissociation is FDR-confirmed, replicates across architectures, and yields a 6/32-layer surgical model at 42% memory. Numerical sensitivity is not functional importance.

7 Limitations

The study covers instruction-tuned Pre-LN transformers with RMSNorm (LLaMA-3.1-8B, Mistral-7B, plus the cross-architecture extensions of Section 4.7). Whether the inversion extends to Post-LN/DeepNorm models, Pre-LN with standard LayerNorm, or base (non-instruction-tuned) models is open. We attempted base-model replication on OPT-6.7B and Pythia-6.9B but zero-shot ARC generation scores below chance (22%, 19%), preventing accuracy-based restoration ranking; loglikelihood scoring via `lm-evaluation-harness` (Gao et al., 2021) is the natural follow-up. The recovery metric is bounded by n_{flipped} per dataset (GSM8K = 31, ARC = 36, MATH = 24), making MATH underpowered. Cross-method replication uses GPTQ; AWQ restoration was infeasible due to library incompatibility on our hardware. Quantization-resistant models (Qwen2.5-Math, Gemma-2) fall outside the probe's operating regime: negligible damage yields a flip-set too sparse for ranking. The HESS-6 baseline ranks layers by the activation-covariance trace alone; deployed GPTQ allocators use the full Hessian (OBS-style update), ResQ uses PCA decomposition, and production systems (SmoothQuant, AWQ-Marlin) combine outlier protection with kernel-fused dequantisation – our comparison isolates the bit-allocation signal, not the full allocator. Identifying the surgical layer set requires the full 64-sub-layer sweep ($\sim 10^4$ generations per candidate model); reported throughput (106 tok/s Surgical vs. 101 tok/s NF4) is single-batch decode on a single A100, not a batched-serving measurement. Bootstrap CIs on ρ and per-task recovery are not yet computed.

Ethical considerations and compute. All models and benchmarks are publicly released artifacts used under their standard research licenses and consistent with their intended research use; no human subjects, annotators, or private data were involved. The probe identifies layers whose precision matters most for specific tasks, which could in principle be misused to target adversarial degradation; we view this risk as low because the same information already enables defensive mixed-precision allocation, which is the framing we adopt here. All experiments (including the 64-sub-layer sweep, multi-seed runs, cross-model replication, GPTQ probe, and Hessian-vs-restoration study) were run on a single NVIDIA A100 (80GB) via Lightning.ai; no training was performed, only inference-time quantization and surgical transplant, so per-condition cost is dominated by generation throughput (Section 3 and Appendix A).

Use of AI assistants. We used a general-purpose AI assistant for code refactoring, and prose polishing of the manuscript. All research questions, experimental design, statistical analysis, scientific claims, and final wording are the responsibility of the human authors; AI-generated text and code were reviewed and edited before inclusion.

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A Multi-Seed Statistical Validation

Figure 5 shows the per-seed accuracy means ($\pm$std), the full McNemar p-value matrix, and the throughput chart that support the headline claims of Section 4.4.

B Full Layer Sensitivity Heatmap

Figure 6 shows the complete 2×3 recovery-rate heatmap: rows = Attention / MLP, columns = GSM8K / ARC / MATH. Each cell shows the layer index and recovery rate $r(\ell, \tau)$. The clean crossover pattern (high recovery in attention-GSM8K and MLP-ARC cells; near-zero recovery in attention-ARC and MLP-GSM8K for top layers) motivates the FDR analysis in Section 4.3.

C Surgical Model Capstone Results

Figure 7 presents the three-panel capstone: accuracy comparison (FP16 / 4-bit / Surgical), percentage of the FP16–4-bit gap recovered per dataset, and memory footprint. The surgical model recovers 70.8% (GSM8K), 52.8% (ARC), 53.2% (MATH) of the respective gaps at 42% of FP16 memory.

D Cross-Model Comparison: Llama-3.1-8B vs. Mistral-7B

Figure 8 shows the full cross-model comparison: capstone accuracy, per-layer attention sensitivity (GSM8K), per-layer MLP sensitivity (ARC), cross-task interference matrix, and summary statistics tables. Both models share critical layer structure (L14 attn, L6–7 MLP), despite having different absolute accuracy levels and quantization behaviour.

E MATH Difficulty Stratification

Figure 9 stratifies MATH examples by difficulty tier (L1–3 = Easy, L4–5 = Hard). Easy tier: FP16–4-bit gap = 3.4pp (p=0.086, *ns*). Hard tier: gap = 1.4pp (p=0.476, *ns*). The *smaller* gap on Hard problems (where both models fail for capability reasons) confirms that MATH is capability-limited rather than precision-limited.

F Bidirectional Flip Analysis

Figure 10 shows the four-quadrant breakdown of all 500 test examples per dataset: both-correct, forward-flipped (FP16$\rightarrow$4-bit, quantization hurts), reverse-flipped (4-bit$\rightarrow$FP16, quantization helps), and both-incorrect. Net forward flips: GSM8K = +31, ARC = +36, MATH = +24. Flipped examples have the same FP16 confidence as stable-correct examples (mean gap = -0.003), ruling out boundary perturbation as an explanation.

G Cross-Quantization-Method Replication (GPTQ): Full Detail

The NF4 restoration sweep assumes that the functional map (which layers are precision-critical for which tasks) reflects a property of the model architecture, not an artifact of NF4's specific quantization mechanism. We test this by running an identical 64-layer restoration sweep on a GPTQ-quantized baseline (Frantar et al., 2023) (hugging-quants/Meta-Llama-3.1-8B-Instruct-GPTQ-INT4, group size 128, `desc_act=True`, Triton v2 kernel, n=500 examples per dataset).

Figure 11 compares NF4 and GPTQ recovery profiles across all 32 layers for attention and MLP components. The key dissociation replicates.

Attention task-selectivity. L14 attention remains the top-ranked layer for GSM8K under GPTQ (46.4% recovery), close to the NF4 value of 51.6%. The differential persists: L14 attention ARC recovery under GPTQ is only 3.6% (1/28 flipped examples), the same near-zero value as under NF4 (8.3%). The attention dead zone (L15–31 for ARC) partially replicates: layers 18, 21, 23, 25, 29–31 all show 0% ARC recovery under GPTQ.

MLP cross-task structure. Under GPTQ, early MLP layers remain cross-task helpers: L6 recovers 46.4% GSM8K and L7 recovers 28.6% ARC, a similar pattern to NF4 (47.9% and 25.0%). A notable shift is that layers L0–L1 MLP become more prominent under GPTQ (35.7% ARC each), likely reflecting GPTQ's calibration redistributing residual error toward earlier layers. The core cross-task property (MLP recovery transfers across tasks in both directions) is preserved.

Interpretation. The replication across fundamentally different quantization algorithms (NF4: scalar round-to-nearest with double quantization; GPTQ: Hessian-calibrated group quantization with activation-order permutation) supports our claim that the functional map is a structural property of the model, not an NF4 artifact. The quantitative shift in recovery magnitudes is expected (GPTQ has different per-layer error distributions), but the

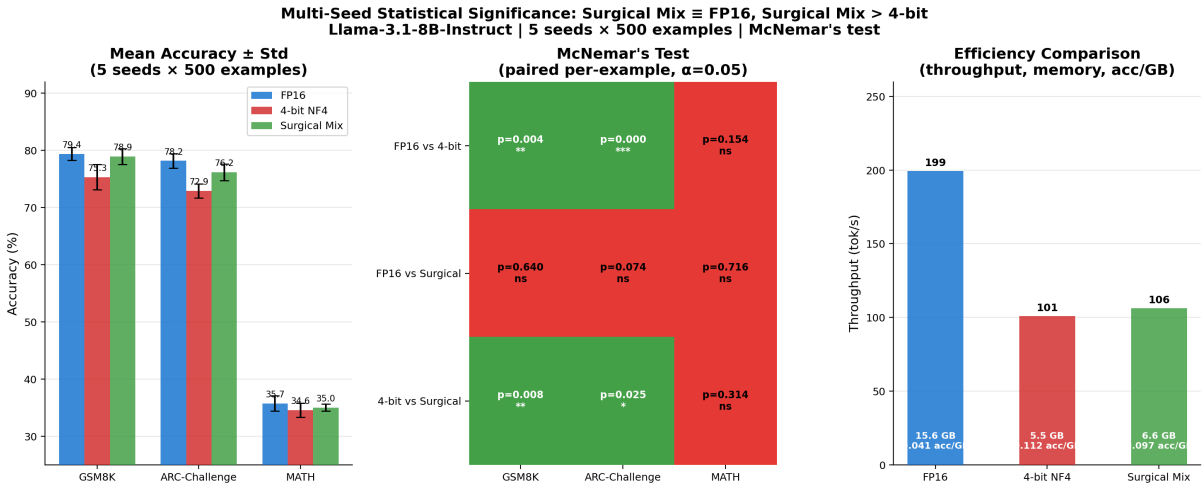


Figure 5: Multi-seed statistical validation (5 seeds, McNemar's test). *Left*: Mean ± std accuracy across seeds. *Centre*: McNemar p-values; green = significant (α=0.05), red = not significant (*ns*). Surgical ≡ FP16 (all *ns*); Surgical > 4-bit (GSM8K p=0.008, ARC p=0.025). *Right*: Throughput (tok/s); surgical matches 4-bit while recovering most of FP16's accuracy. All Cohen's $d \leq 0.12$.

qualitative dissociation (L14 attention arithmetic-selective; MLP L6–7 cross-task) is stable.

H Extended Multi-Model Probe: Full Detail

We extend the restoration probe to three additional models: DeepSeek-R1-Distill (DeepSeek-AI et al., 2025) (Llama-8B distillation of a large reasoning model), Qwen2.5-Math-7B-Instruct (Yang et al., 2024) (a math-specialised model), and Gemma-2-9B-IT (Gemma Team et al., 2024). The results reveal two qualitatively distinct behaviours.

Replication (DeepSeek-R1). DeepSeek-R1-Distill suffers 2.4pp GSM8K and 5.0pp MATH NF4 damage (FP16: 64.8%/46.6%; NF4: 62.4%/41.6%). The restoration sweep confirms L13 attention as the top-ranked layer for GSM8K (recovery rate = 42.6%), directly replicating the L13–14 arithmetic-attention finding on a third architecture. MLP L6 also replicates for commonsense recovery (33.3%).

Quantization-resistant models (Qwen2.5-Math, Gemma-2). Qwen2.5-Math-7B exhibits near-zero NF4 damage on all tasks ($\leq$1.0pp). Gemma-2-9B similarly shows no significant degradation (GSM8K: $-$0.8pp; ARC: +0.8pp; MATH: 0.0pp). With minimal quantization damage there is no recovery signal to probe; the sparse flip sets (n<10) yield noisy recovery estimates unsuitable for ranking. This *quantization resistance* is itself a finding: model architecture and training regime affect precision sensitivity, and the restoration probe correctly diagnoses the absence of functional degradation rather than producing spurious rankings. Together, the three-model extension shows that L13–14 arithmetic-attention criticality is a property shared across architecturally diverse models that exhibit quantization damage, while quantization-robust models fall outside the probe's operating regime.

I Mistral Hessian-vs-Restoration Replication

We rerun the Phase 21 ranking comparison on Mistral-7B using the same methodology as on Llama-3.1-8B: 128 WikiText-2 calibration sequences for the activation-covariance Hessian proxy, and per-sub-layer restoration recovery rates from the cross-task sweep already reported in Section 4.7. Across all 64 Mistral-7B sublayers (32 attention, 32 MLP), the Spearman correlation between Hessian rank and restoration rank is $\rho = -0.574$ (p<0.001, n=64), with per-task breakdowns of $\rho_{\text{GSM8K}} = -0.458$ (p<0.001), $\rho_{\text{ARC}} = -0.536$ (p<0.001), and $\rho_{\text{MATH}} = -0.342$ (p=0.006). All three task-specific correlations are negative and significant, mirroring the Llama-3.1-8B pattern in which ARC shows the strongest inversion. The aggregate magnitude on Mistral-7B ($|\rho|$=0.574) is in fact larger than on Llama-3.1-8B ($|\rho|$=0.506), despite Mistral-7B's grouped-query attention and substantially different baseline accuracy, confirming that the Hessian–functional decou-

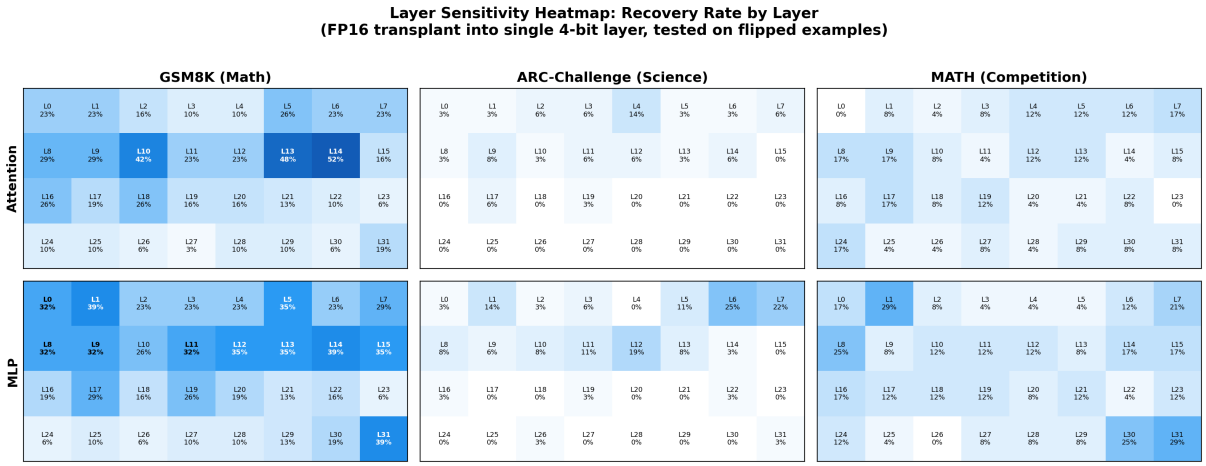


Figure 6: Full 64-cell layer sensitivity heatmap. Rows = Attention (top), MLP (bottom); Columns = GSM8K, ARC-Challenge, MATH. Colour intensity = recovery rate (white→dark blue = 0→55%). Layer indices and recovery percentages shown in each cell. Critical cells highlighted by high recovery values: L13–14 (Attention/GSM8K), L6–7 (MLP/ARC).

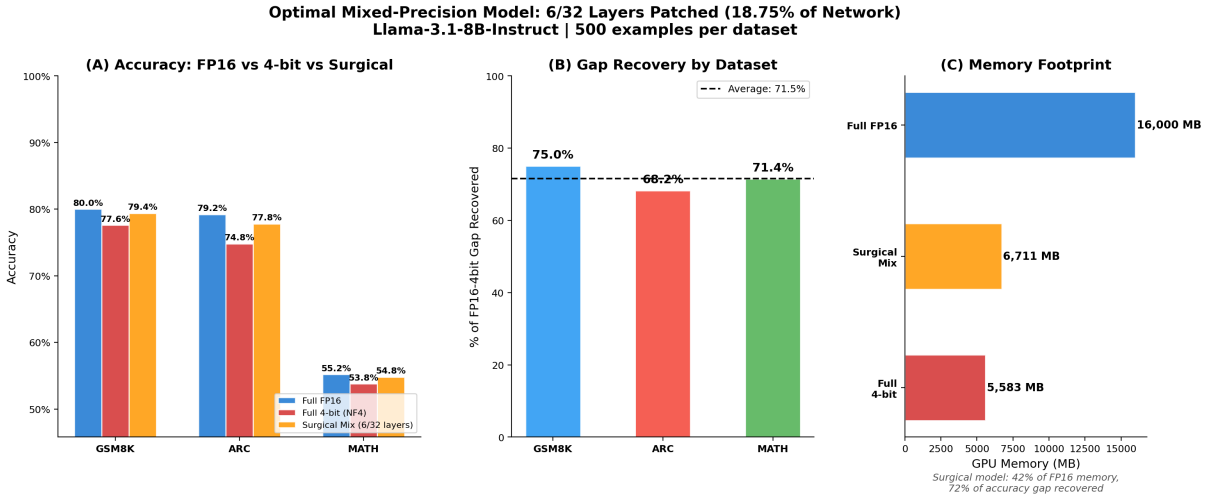


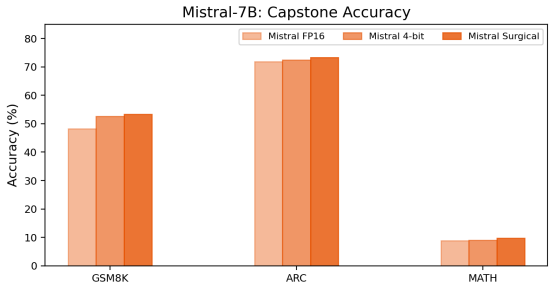
Figure 7: Optimal mixed-precision capstone. *Left*: Absolute accuracy for FP16, 4-bit, and Surgical (6/32 layers). *Centre*: Percentage of FP16–4-bit gap recovered per dataset. *Right*: GPU memory footprint. Surgical model memory (6.71 GB) is only 1.13 GB above pure 4-bit (5.58 GB).

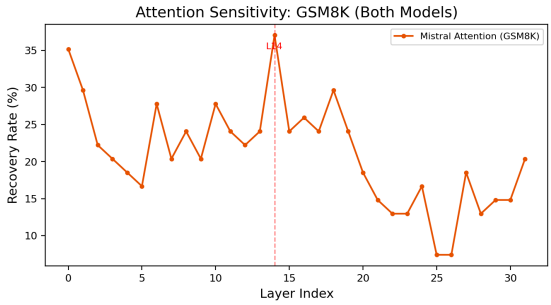
pling is not a Llama-3.1-8B-specific artifact but a structural property of Pre-LN transformers under 4-bit quantization.

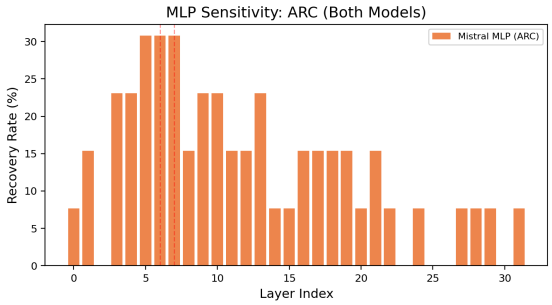
J Visualisation of HESS-6 vs. REST-6 (Phase 24)

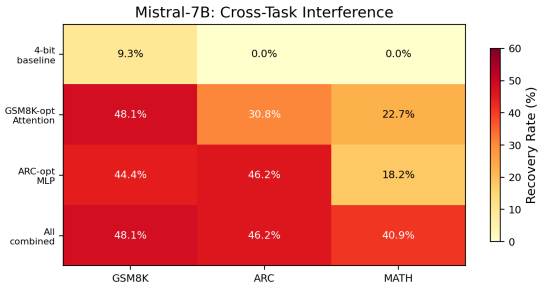
Figure 12 visualises the per-condition accuracies and recovery rates of Table 3. All numerical values shown are reproduced in the table; this figure is provided for visual comparison.

Plot 12: Cross-Model Replication — Llama-3.1-8B vs Mistral-7B









Mistral-7B: McNemar's Test (3 seeds, n=1500)

	GSM8K	ARC	MATH
FP16 mean	50.9%	70.5%	10.4%
4-bit mean	53.0%	71.5%	9.7%
Surgical mean	53.8%	71.7%	9.5%
FP16 vs 4-bit	p=0.116 ns	p=0.129 ns	p=0.439 ns
Surg vs FP16	p=0.033 ✓	p=0.030 ✓	p=0.288 ns
Surg vs 4-bit	p=0.523 ns	p=0.617 ns	p=0.857 ns

Cross-Model Comparison Summary

Finding	Llama-3.1-8B	Mistral-7B
Critical attn layer	L13, L14	L14, L0
Critical MLP layers	L1, L6, L7, L31	L3, L5, L6, L7
Quantization effect	Damages (−2.4pp GSM8K)	Neutral (ns)
Surgical vs FP16	Equivalent (ns)	Surgical > FP16 (p<0.05)
Surgical vs 4-bit	Surgical > 4-bit (p<0.01)	Equivalent (ns)
ARC dead zone	L15–31	L23–31
MATH result	Null (capability)	Null (capability)
Attn monosemantic?	Yes	Yes
MLP superposition?	Yes	Yes

Figure 8: Cross-model replication: Llama-3.1-8B (blue) vs. Mistral-7B (orange). *Top row*: Capstone accuracy and per-layer attention sensitivity. *Middle row*: MLP sensitivity and cross-task interference matrix. *Bottom row*: McNemar significance tables for both models. Both architectures identify L14 attention and L6–7 MLP as critical layers.

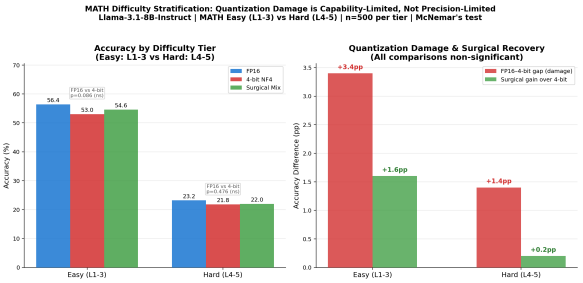


Figure 9: MATH difficulty stratification. All comparisons are non-significant (McNemar's test, n=500 per tier). Hard problems show *smaller* FP16–4-bit gap than Easy problems, because both models fail hard problems for capability reasons, not precision.

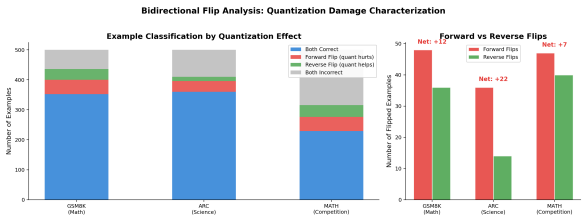


Figure 10: Bidirectional flip analysis. *Left*: Stacked bar showing four-quadrant breakdown per dataset. *Right*: Forward vs. reverse flip counts with net annotations. Forward-flipped examples have the same FP16 confidence as stable-correct examples, confirming systematic (not boundary) corruption.

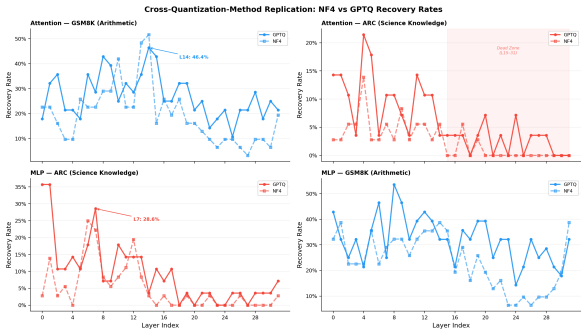


Figure 11: NF4 (dashed) vs. GPTQ (solid) recovery rates for all 32 layers. *Top*: Attention; *Bottom*: MLP. Left panels = GSM8K (arithmetic); Right panels = ARC (science knowledge). L14 attention remains the top-ranked layer for arithmetic in both methods (46.4% GPTQ vs. 51.6% NF4). The ARC dead zone (L15–31, shaded) partially replicates under GPTQ.

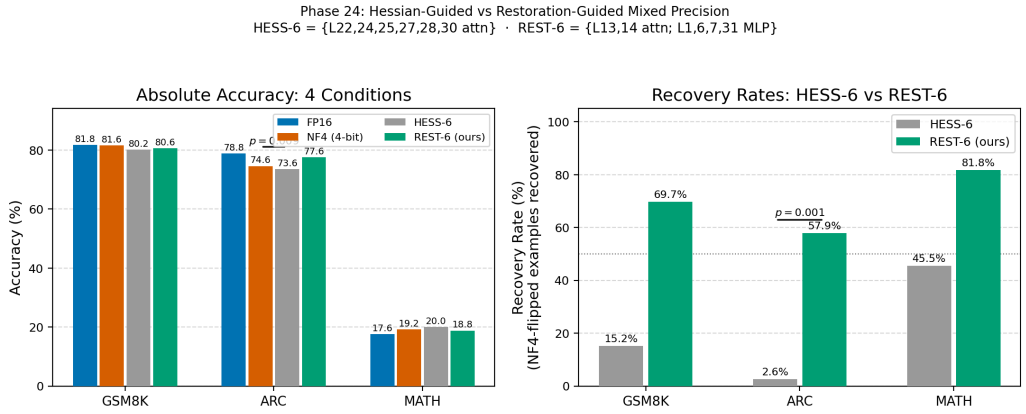


Figure 12: HESS-6 (grey) vs. REST-6 (green) at matched 6/32 FP16 budget. *Left*: absolute accuracy; *Right*: NF4-damage recovery rates. On ARC, REST-6 recovers 52.6% (p=0.009 vs. NF4) vs. HESS-6 2.6% (p=0.125, ns); direct comparison p=0.001.